CreditGuard AI - Solution Architecture

From raw portfolio data to governed analyst decisions

01 Source data

Five banking tables

- Customers and loans
- Bureau credit lines
- Repayments and collections

CSV | pandas

02 Data foundation

Clean account-level view

- Validated one-to-many joins
- Quality flags and normalization
- Parquet dataset + QA report

pandas | PyArrow

03 Risk intelligence

Scores and explanations

- Logistic baseline
- Gradient boosting challenger
- AUC, KS, Gini + importance

scikit-learn

04 Analyst experience

Monitor, explore, review

- Portfolio and account views
- Filters and drill-down
- Governance metrics

Streamlit | Plotly

Embedded AI with explicit guardrails

Ask the Portfolio

Pandas calculates an approved aggregation first. OpenRouter only rewrites the verified fact.

Question + aggregate fact only

Deterministic fallback

Anomaly review

Peer comparison flags unusual behaviour and gives the analyst a plain-language reason code.

Analyst confirms or dismisses

No automated adverse action